

Deploying DevOps Automation Agents



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What Is Azure AI Foundry?

Unified development platform for AI applications on Azure

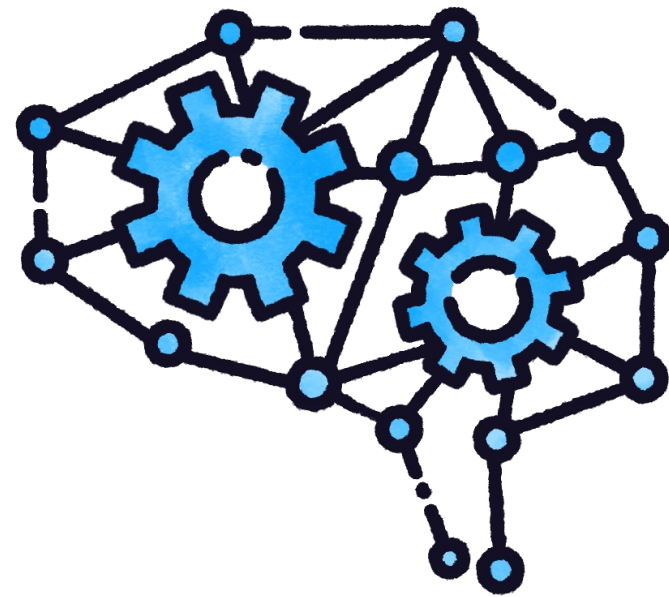
Web-based interface for managing models, projects, and resources

Simplifies deployment and monitoring of AI applications

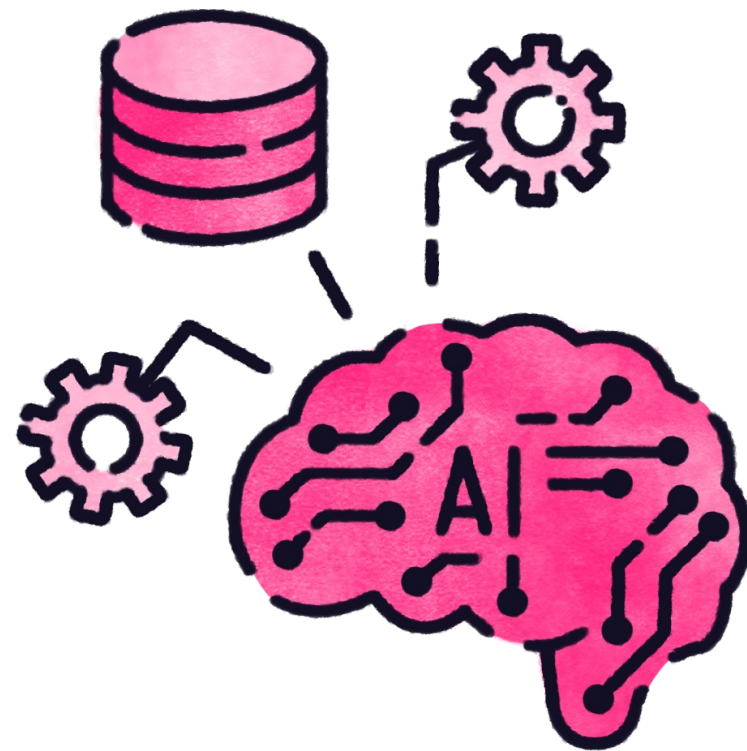
Bundles required Azure services automatically into integrated workspaces



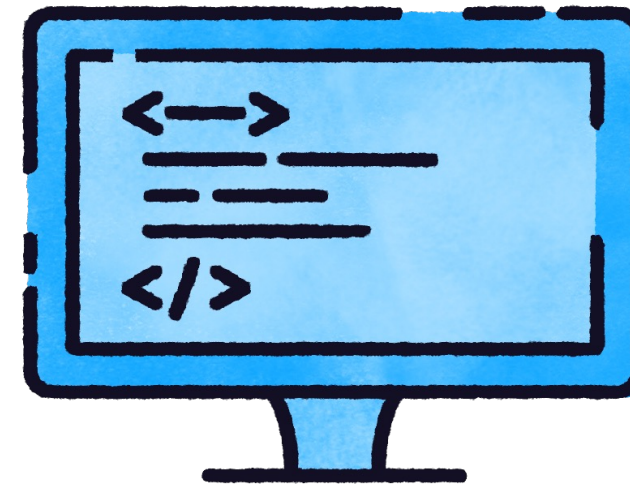
Azure AI Agents Fundamentals



**Built on Azure
OpenAI Service**



**Configured
through Azure AI
Foundry**



**Native Azure
DevOps
integrations**



**Built-in security
and audit
logging**



Python SDK Integration

**Programmatic access to
Azure AI Foundry**

**Inherits enterprise
security and Entra
Agent ID authentication**

**Maintains agent context
and tool integrations
from Foundry**

**Built-in Azure credential
handling**

**Built-in compliance
logging**





Demo: Building an Azure AI Agent

Set up an Azure AI Project, deploy a GPT-4o model, and create a Python application that connects to our Azure AI agent



Building a Specialized Log Analysis Agent



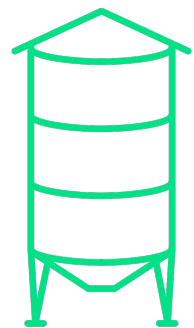
DevOps Workflow Failure Patterns



Teams encounter recurring workflow failures



Manual log analysis wastes developer time



Troubleshooting knowledge is often siloed



CI/CD growth outpaces troubleshooting capacity



AI Agents for DevOps



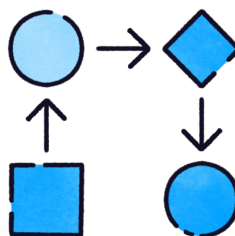
AI recognizes failure patterns instantly



Specialized prompts encode troubleshooting expertise



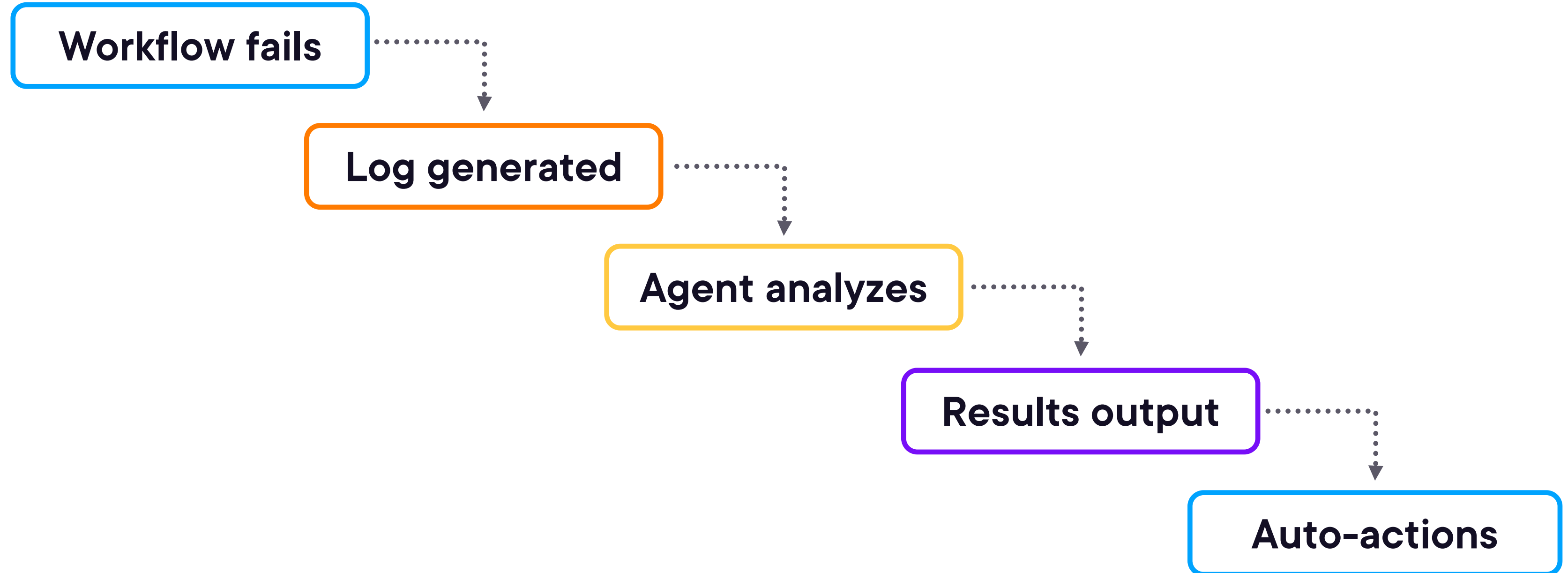
Agents democratize senior-level knowledge



Automated analysis scales troubleshooting capacity



DevOps Agent Implementation Strategy



Demo: Building and Testing a Log Analysis Agent

Build a specialized Python-based AI agent that analyzes GitHub Actions workflow logs, identifies root causes, and recommends solutions



Building a GitHub Workflow Intelligence Agent



Demo: Creating a GitHub Intelligence Azure AI Agent via the Browser

Build an Azure AI Agent via the Azure Portal



Demo: Adding Tools to our Azure AI Agent

Adding Github issues and Github Actions tools

